LED Behavior During BLE Pairing

3 LED Colors: White (neutral) · Green (success) · Red (error)

HARDWARE

Front-mounted LED strip (horizontal bar). 3 colors available: **White**, **Green**, **Red**. Patterns: Solid ON, OFF, Slow blink, Fast blink, Double-blink.

Color	Meaning	Used In
WHITE	Neutral / in-progress / waiting	States 1-5 (discovery → verification)
GREEN	Success / connected / ready	States 6-8 (paired, WiFi, complete)
RED	Error / emergency / stop	States 9-10 (failed, E-Stop)

#	State	Color	Pattern	Timing	Meaning
1	**Discoverable**	**WHITE**	Slow blink	1s on / 1s off	I'm here. Ready to pair.
2	**Found**	**WHITE**	Fast blink	300ms on / 300ms off	I see your phone.
3	**Pairing**	**WHITE**	Double blink	100ms on, 100ms off, 100ms on, 700ms off	Connecting. Please wait.
4	**Awaiting Code**	**WHITE**	Solid ON	Constant	Enter the code. Your turn.
5	**Verifying**	**WHITE**	Rapid blink	100ms on / 100ms off	Checking code...
6	**Success**	**GREEN**	3 flashes → solid	150ms on/off ×3 then constant	Paired! Ready for Wi-Fi.
7	**Wi-Fi Setup**	**GREEN**	Slow blink	2s on / 0.5s off	Connecting to Wi-Fi...
8	**Complete**	**GREEN**	Solid ON	Constant	All set. Ready to work.
9	**Failed**	**RED**	3 flashes → off	500ms on/off ×3 then → State 1	Error. Try again.
10	**E-STOP**	**RED**	**Rapid flash**	**200ms on / 200ms off NON-STOP**	**EMERGENCY. Stopped.**

STATE FLOW

```
POWER ON → [1.White Blink] → [2.White Fast] → [3.White Double] → [4.White Solid] → [5.White Rapid]
■■ OK → [6.Green Flash] → [7.Green Blink] → [8.Green Solid] → READY
■■ FAIL → [9.Red Flash] → back to [1]      [10.Red Rapid] = E-STOP, overrides ALL
```

NOTES

- **White** = robot is working/waiting. **Green** = something succeeded. **Red** = something went wrong or emergency.
- E-Stop (#10) is HIGHEST PRIORITY — interrupts any pattern immediately. Red rapid flash until cleared.
- State 1 is default after power-on. Robot returns to it after failure or 2-min timeout.
- Transition from White→Green happens at State 6 (pairing success). This is the key visual "it worked" moment.
- Blink timings are approximate — ±50ms acceptable. Pattern distinctness matters more than exact timing.
- Hardware coordination needed with Sufyan for LED strip placement and wiring.